Mohammad Zhalechian

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APPOINTMENT

Indiana University, Bloomington, IN *2023 – Present*
Assistant Professor of Operations and Decision Technologies
Kelley School of Business

Harvard University, Cambridge, MA *2022 – 2023*
Postdoctoral Fellow
Harvard Kennedy School

EDUCATION

University of Michigan, Ann Arbor, MI
Ph.D. in Industrial and Operations Engineering (Operations Research) *2022*
M.A. in Statistics *2019*

University of Tehran, Tehran, Iran
M.Sc. in Industrial Engineering and Operations Research *2017*

K. N. Toosi University of Technology (KNTU), Tehran, Iran
B.Sc. in Industrial Engineering and Operations Research *2014*

RESEARCH INTERESTS

- Sequential and Data-Driven Decision-Making
- Design and Evaluation of Human-AI Systems
- Algorithm Applications in Healthcare and High-Stakes Settings

PUBLICATIONS

[† industry collaborators; (M.D.) M.D. collaborators]

Published Papers

[1] **Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy**
Management Science, 2026
M. Zhalechian, S. Saghafian, O. Robles†

- *Media coverage (Harvard Kennedy School):* A Human-Algorithm Approach to Approving Medical Devices Could Slash the Recall Rate and Save Billions
- *Media coverage (INFORMS press release):* AI Could Cut Medical Device Recalls by Nearly a Third
- *Media coverage (Medical Xpress):* AI Could Cut Medical Device Recalls by Nearly a Third
- *Media coverage (Today's Medical Developments):* Artificial Intelligence Could Reduce Medical Device Recalls
- *Media coverage (Devdiscourse):* Transforming FDA Clearance: How AI and Human Insight Can Improve Medical Device Safety

[2] **Contextual Learning with Online Convex Optimization with Applications to Medical Decision-Making**
Management Science, 2025
Collaboration with Massachusetts Hospital
E. Keyvanshokooh, M. Zhalechian, C. Shi, MP. Van Oyen, P. Kazemian

- Finalist, INFORMS Minority Issues Forum (MIF) Paper Competition, 2026
- Finalist, POMS College of Healthcare Operations (CHOM) Best Paper Competition, 2022
- Finalist, INFORMS MSOM Best Student Paper Competition, 2021
- Finalist, INFORMS Health Applications Society (HAS) Best Student Paper Competition, 2021
- Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper, 2020

[3] **Dynamic Care Unit Placements under Unknown Demand with Learning**
Manufacturing & Service Operations Management, 2025
Collaboration with St. Joseph Hospital
A. Dean, M. Zhalechian, MP. Van Oyen

[4] **Data-driven Hospital Admission Control: A Learning Approach**
Operations Research, 2023
Collaboration with St. Joseph Hospital
M. Zhalechian, E. Keyvanshokooh, C. Shi, MP. Van Oyen

- Winner, IOE Richard C. Wilson Best Student Paper Award on Service Systems, 2021
- Finalist, Seth Bonder Scholarship, INFORMS Health Applications Society, 2020

[5] **Online Resource Allocation with Personalized Learning**
Operations Research, 2022
Collaboration with Michigan Medicine
M. Zhalechian, E. Keyvanshokooh, C. Shi, MP. Van Oyen

[6] **Validation of a Visual Field Prediction Tool for Glaucoma: A Multicenter Study Involving Patients with Glaucoma from Throughout the United Kingdom**
American Journal of Ophthalmology, 2025
Collaboration with Medisoft UK
A. Dean, DJ. Fu (M.D.), M. Zhalechian, MP. Van Oyen, M. Lavieri, AP. Khawaja (M.D.), J. Stein (M.D.)

[7] **Augmenting Kalman Filter Machine Learning Models with Data from Optical Coherence Tomography to Predict Future Visual Field Loss: The African Descent and Glaucoma Evaluation Study**
Ophthalmology Science, 2022
Collaboration with Kellogg Eye Center and Shiley Eye Institute
M. Zhalechian, MP. Van Oyen, M. Lavieri, C. Andrew†, J. Stein (M.D.), et al.

Working Papers

[8] **From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement**
Under Review
Collaboration with Michigan Medicine
A. Dean, M. Zhalechian, I. Attari, M. Garifullin†, J. Pardo†, MP. Van Oyen
- Implemented at Michigan Medicine since July 2023
- Finalist, INFORMS Service Science Cluster Best Paper Competition, 2026
- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026
- Accepted at Michigan Medicine Quality Month Symposium, 2024

[9] **Does Early Pain Management Pay Off? Evidence on Utilization, Costs, and Surgeries**
Under Review
Collaboration with Integrative Sports & Spine Pain Clinic in Southern California
B. Li, M. Zhalechian, C. Chen, N. Vaughn (M.D.), H. Lee (M.D.)

[10] **Learning Bundle Pricing of Reusable Resources**
Under Review
A. Dean, M. Zhalechian, C. Shi
- Finalist, Service Science Best Student Paper Competition, 2024

[11] **Operations Management for Restorative Public Health: Grand Challenges in Reversing the Damage of the Opioid Crisis**
Major Revision at ***Manufacturing & Service Operations Management***
C. Chen, J. Helm, B. Li, J. Mejia, M. Zhalechian (alphabetical authors)
- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026

Selected Published Papers Prior to Ph.D.

Sustainable Design of a Closed-loop Location-routing-inventory Supply Chain Network Under Mixed Uncertainty
Transportation Research Part E, 2016
M. Zhalechian, R. Tavakkoli-Moghaddam, B. Zahiri, M. Mohammadi

Hub-and-spoke Network Design under Operational and Disruption Risks
Transportation Research Part E, 2017
M. Zhalechian, SA. Torabi, M. Mohammadi

A Self-adaptive Evolutionary Algorithm for a Fuzzy Multi-objective Hub Location Problem: An Integration of Responsiveness and Social Responsibility
Engineering Applications of Artificial Intelligence, 2017
M. Zhalechian, R. Tavakkoli-Moghaddam, Y. Rahimi

HONORS AND AWARDS

(∗ denotes honors or awards received by students)

2026 Finalist, INFORMS Minority Issues Forum (MIF) Early Career Award

2026 Finalist, INFORMS Junior Faculty Interest Group (JFIG) Teaching Excellence Award

2026 Finalist, INFORMS Data Mining Society Teaching Award

2026 Finalist, INFORMS Service Science Cluster Best Paper Competition

2026 Finalist, INFORMS Minority Issues Forum (MIF) Paper Competition

2025 Operations and Decision Technologies Faculty Scholar for Research Excellence

2025 Kelley School of Business Research Award

2024* Finalist, Service Science Best Student Paper Competition

2022 Towner Prize for Distinguished Academic Achievement, University of Michigan

2022 Rackham Outstanding Graduate Student Instructor Award, University of Michigan

2022 Finalist, POMS College of Healthcare Operations (CHOM) Best Paper Competition

2021 Rackham Graduate Research Grant, University of Michigan

2021 Winner, IOE Richard C. Wilson Best Student Paper Award, University of Michigan

2020 Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper

2020 Finalist, INFORMS Seth Bonder Scholarship for Applied OR in Health Services

2020 IOE Brown Outstanding Graduate Student Instructor Award, University of Michigan

2018–2021 Rackham Travel Grant Awards, University of Michigan

2017 Graduate Student of the Year, College of Engineering, University of Tehran

INVITED SEMINARS AND CONFERENCE PRESENTATIONS

From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement

- MSOM Conference, Special Interests Group *Jul. 2026*

Interpretable Design of Multi-Agent Mobile Health Applications

- Center for the Business of Life Sciences, Kelley School of Business *Oct. 2026*
- Workshop on AI & Analytics for Social Good, College Park, MD *Apr. 2026*
- INFORMS Healthcare Conference, Toronto, Canada *Jul. 2023*

Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy

- Wayne State University, Industrial & Systems Engineering Seminar Series *Nov. 2025*
- INFORMS Annual Meeting (2023, 2024, 2025); MSOM Conference (2025); POMS Annual Conference (2024); INFORMS Healthcare Conference (2023)

Data-Driven Learning and Resource Allocation in Service Operations Management

- Dartmouth College, Tuck School of Business *Jan. 2023*
- Indiana University, Kelley School of Business *Jan. 2023*
- Carnegie Mellon University, Tepper School of Business *Dec. 2022*
- Rice University, Jones Graduate School of Business *Dec. 2022*
- University of North Carolina at Chapel Hill, Kenan-Flagler Business School *Dec. 2022*
- University of Illinois Urbana-Champaign, Gies College of Business *Dec. 2022*
- Miami University, Herbert Business School *Dec. 2022*

- University of Illinois Chicago, College of Business Administration *Dec. 2022*
- Baruch College, Zicklin School of Business *Dec. 2022*

Personalized Hospital Admission Control: A Contextual Learning Approach

- INFORMS Annual Meeting (2019, 2020, 2021, 2022); POMS Annual Conference (2019, 2021, 2022); INFORMS Healthcare Conference (2019)

Online Advance Appointment Scheduling with Personalized Learning

- INFORMS Annual Meeting (2020, 2021)

Joint Learning and Optimization for Healthcare Operations and Medical Decision-Making

- INFORMS Annual Meeting, Seth Bonder Scholarship Session *Nov. 2020*

A Distributionally Robust Capacity Planning Model for Optimizing Access Delay in Surgical Services under Limited Information

- INFORMS Healthcare Conference (2019); INFORMS Annual Meeting (2018)

TEACHING EXPERIENCE

Instructor, Indiana University, Bloomington, IN *Spring 2024, 2025 & 2026*
Class: *Business Analytics and Modeling* (K353)
Three sections per year
Instructor rating: 6.5/7 average across nine sections; 6.6/7 most recent (program mean: 6.2)

Instructor, University of Michigan, Ann Arbor, MI *Fall 2020*
Class: *Economic Decision Making* (IOE 201)
Instructor rating: 4.9/5 (college median: 4.5)

Recitation Instructor, University of Michigan, Ann Arbor, MI *Winter 2020*
Class: *Introduction to Markov Processes* (IOE 316)

Teaching Assistant, University of Michigan, Ann Arbor, MI

- Practicum in Hospital Systems (IOE 481) *Fall 2021 – Winter 2022*
- Service Operations Management (IOE 419) *Winter 2021*
- Probability and Statistics for Eng. (IOE 265) *Fall 2018, Winter 2019, Fall 2019*

ADVISING AND MENTORING

PhD Student Committees

- Ali Kaan Kurbanzade, Kelley School of Business, 2025–present
- Iman Attari, Kelley School of Business, 2023–2025
  Placement: University of Cincinnati (Lindner College of Business)

Master's and Undergraduate Mentoring

- Faculty Mentor for Master's Students, Faculty Assistance in Data Science (FADS) Program, Kelley School of Business
- Faculty Mentor for Undergraduate Students, Center of Excellence for Women & Technology, Indiana University

ACADEMIC SERVICE

Service to the Profession

- *Referee:* Management Science, Operations Research, Manufacturing & Service Operations Management, Production and Operations Management, Information Systems Research, European Journal of Operational Research, Omega, Health Care Management Science
- *Judge:* Public Sector OR Best Paper Competition, INFORMS, 2023

Session Organizer & Chair for Conferences

2026 INFORMS Annual Meeting (AI in Healthcare); INFORMS Healthcare Conference (AI for Healthcare Oversight and Operations & AI and Analytics for Decision-Making in Healthcare)

2025 INFORMS Annual Meeting (Data-driven Healthcare Analytics)

2024 POMS Annual Conference (Policies and Regulation in Healthcare)

2021 INFORMS Annual Meeting (Sequential Decision Making and Machine Learning in Healthcare); POMS Annual Conference (Healthcare Operations Management)

2020 INFORMS Annual Meeting (Predictive Analytics & Sequential Decision-Making in Healthcare)

Department and School Service

- PhD Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2025
- Faculty Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2024
- Faculty Recruiting Committee (Graduate Student Committee), Industrial and Operations Engineering, University of Michigan, 2021
- Chair, Anti-Racism and DEI Committee, INFORMS Student Chapter, University of Michigan, 2020–2021
- Chair, Professional Development Committee, INFORMS Student Chapter, University of Michigan, 2019–2020